

AI X CITY CLIMATE ACTION HACKATHON 2026

FloodFact AI

Live Product Demonstration

Every screen in this file is a real screenshot of the running application, captured with Playwright — not a mockup. Repository: github.com/VinceOdundo/FloodfactAI

Two independent layers of realism, both honestly labeled

Whole-app demo mode

No Supabase project is configured for this deck. DEMO_MODE drives every screen from the same seed narrative as supabase/seed.sql — the identical query functions that read live Postgres through RLS in production instead read static fixtures.

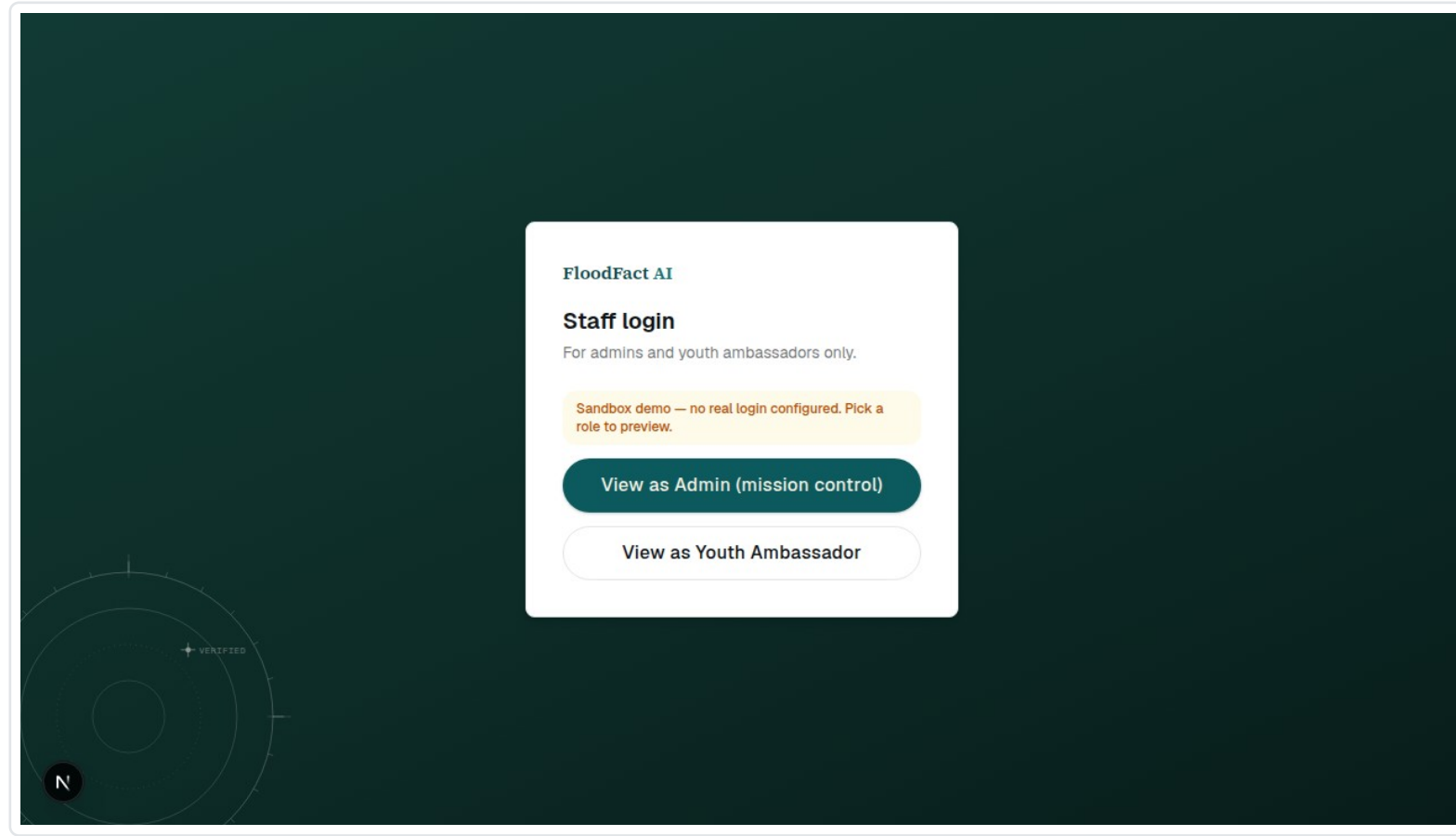
Per-provider sandbox mode

Independently, each external integration — WhatsApp, SMS, the LLM, embeddings, flood-risk — self-selects live or sandbox based on whether its own credential is present. Sandbox evidence is always stamped quality: "degraded," never "ok."

One deploy away from real

Add Supabase credentials and fixtures become live rows. Add one provider's API key and that one adapter goes live. Zero code changes either way — see docs/SETUP.md.

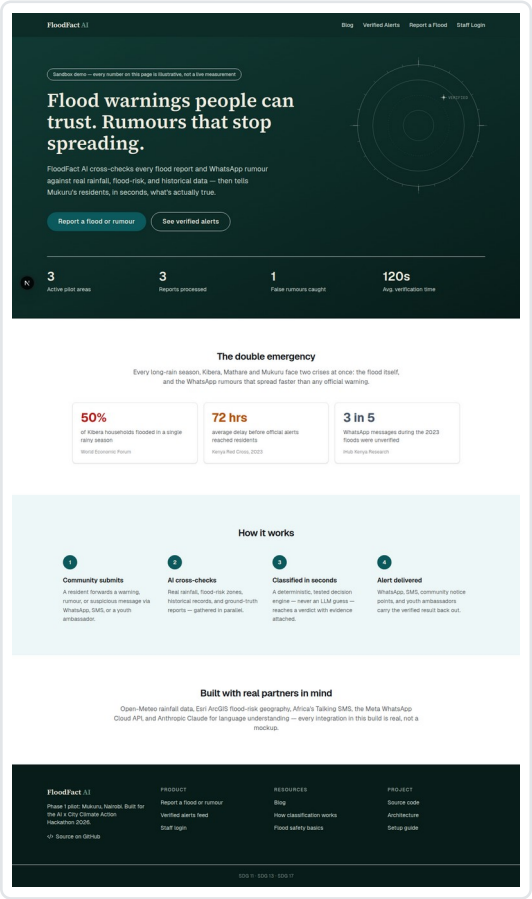
One login screen swaps in two real roles



Sandbox mode replaces Supabase Auth with a role switcher — Admin or Youth Ambassador — no accounts needed.

STEP 1 — A RESIDENT ARRIVES

The public site states the deal in one screen



No install, no account — the landing page links straight into reporting and the public alerts feed.

STEP 2 — REPORTING TAKES SECONDS

“Water is rising fast near the railway crossing on Kanini Road”

FloodFact AI

[Blog](#)[Verified Alerts](#)[Report a Flood](#)[Staff Login](#)

Report a flood or a suspicious message

Tell us what you're seeing, or paste a message you're unsure about. We'll check it against current rainfall, flood-risk maps and historical records, and reply with a clear verdict — Verified Warning, Elevated Risk, or False Information.

What are you seeing or hearing? *

Water is rising fast near the railway crossing on Kanini Road

Where? (street, landmark, or sub-area)

Kanini Road, Mukuru kwa Reuben

Phone number (optional — to receive the verified result directly)

+254712345678

Submit report

Prefer WhatsApp or SMS? This same system checks messages sent there too — this form is here for anyone who can't.

FloodFact AI

Phase 1 pilot: Mukuru, Nairobi. Built for the AI x City Climate Action Hackathon 2026.

🔗 Source on GitHub

PRODUCT

[Report a flood or rumour](#)

[Verified alerts feed](#)

[Staff login](#)

RESOURCES

[Blog](#)

[How classification works](#)

[Flood safety basics](#)

PROJECT

[Source code](#)

[Architecture](#)

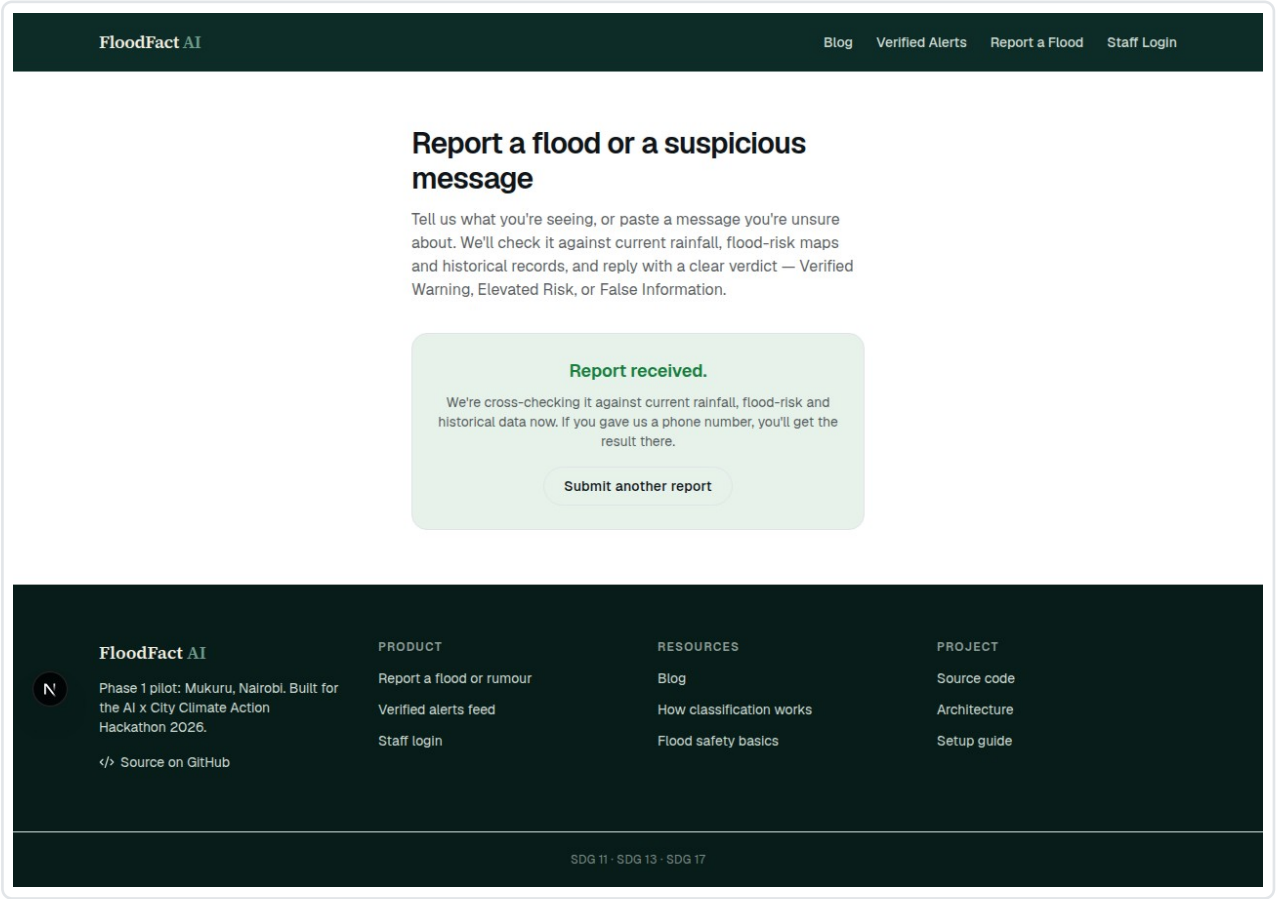
[Setup guide](#)

SDG 11 · SDG 13 · SDG 17

This same Zod-validated schema also accepts inbound WhatsApp and SMS — the form is the fallback channel.

STEP 3 — IMMEDIATE ACKNOWLEDGMENT

The resident is told their report is already moving



Submission triggers the classification pipeline immediately — no queue, no wondering if it arrived.

Every report runs the same six-stage, fully-audited pipeline

This is the real code path — `lib/pipeline/classify.ts` calling `lib/core/risk-engine.ts` — not a simplified diagram.

1

Understand

Claude extracts hazard type, claimed location, and urgency — logged verbatim to `nlu_extractions`.

2

Geocode

The claimed location is matched against a real Mukuru gazetteer and resolved to a pilot area via PostGIS.

3

Gather Evidence

Rainfall, flood-risk zone, historical base rate, corroboration, ambassador ground-truth, and rumor similarity — in parallel.

4

Decide

`risk-engine.ts` scores the evidence. Pure, deterministic, 23 unit tests. No LLM call in this step.

5

Explain

Claude writes the plain-language, bilingual EN/Swahili rationale for a verdict it did not make.

6

Alert

A channel-specific message goes out over WhatsApp, SMS, the public feed, and the ambassador queue.

STEP 4 — INSIDE A SINGLE CASE

A verdict that shows its work

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Overview

Escalations

Pilot Metrics

GIS Map

Data Sources

Ambassadors

MISSION CONTROL

Ops Admin (demo) Sign out

← Back to overview

REPORT 30000000

Mukuru kwa Reuben

Water is rising fast near the railway crossing in kwa Reuben, already at knee height on Kanini Road

Channel: whatsapp Hazard type: Flood Status: classified 9/1/2026, 11:31:15 AM

Verified Warning

Confidence: 88%

Heavy rainfall in the last 3 hours, the location sits inside a mapped high flood-risk zone, similar events have happened here before, and 2 other nearby reports corroborate rising water.

EVIDENCE SNAPSHOT

SOURCE	QUALITY	VALUE	FETCHED
open meteo	Degraded	24.5	11:31:15 AM
arcgis flood risk	Degraded	1	11:31:15 AM

ALERT DELIVERY

VERIFIED FLOOD WARNING – Mukuru kwa Reuben. Water is rising. Move to higher ground now and avoid low-lying paths and river crossings. Verified by FloodFact AI. Evidence: Heavy rainfall in the last 3 hours, the location sits inside a mapped high flood-risk zone, similar events have happened here before, and 2 other nearby reports corroborate rising water. Issued 14:31.

ambassador_queue: sent

REPORTER CONTACT

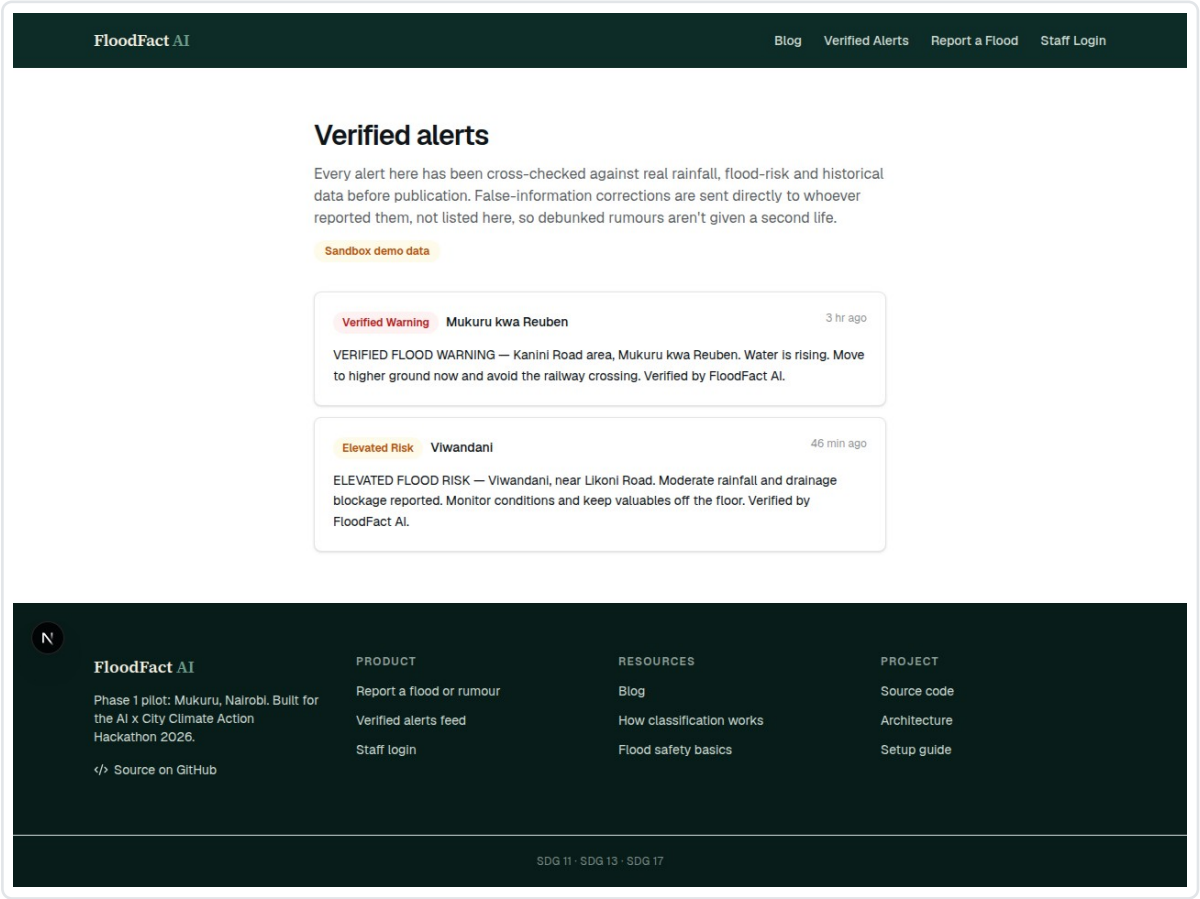
Phone numbers are stored separately with no default read access. Revealing one is logged to the audit trail.

Reveal contact (audited)

Verified Warning, a confidence score, a written rationale, and the evidence_items row behind each one.

STEP 5 — THE VERDICT REACHES THE PUBLIC

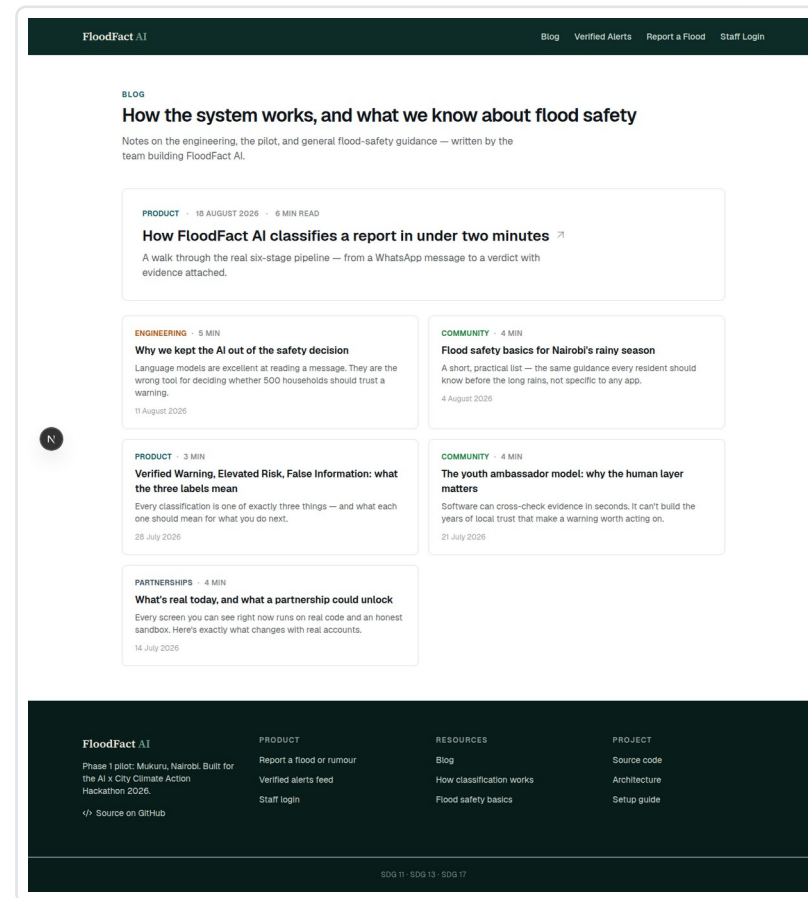
Only verified and elevated-risk classifications appear here



Reads from a dedicated public_alert_feed view — no raw reports, contact details, or open cases exposed.

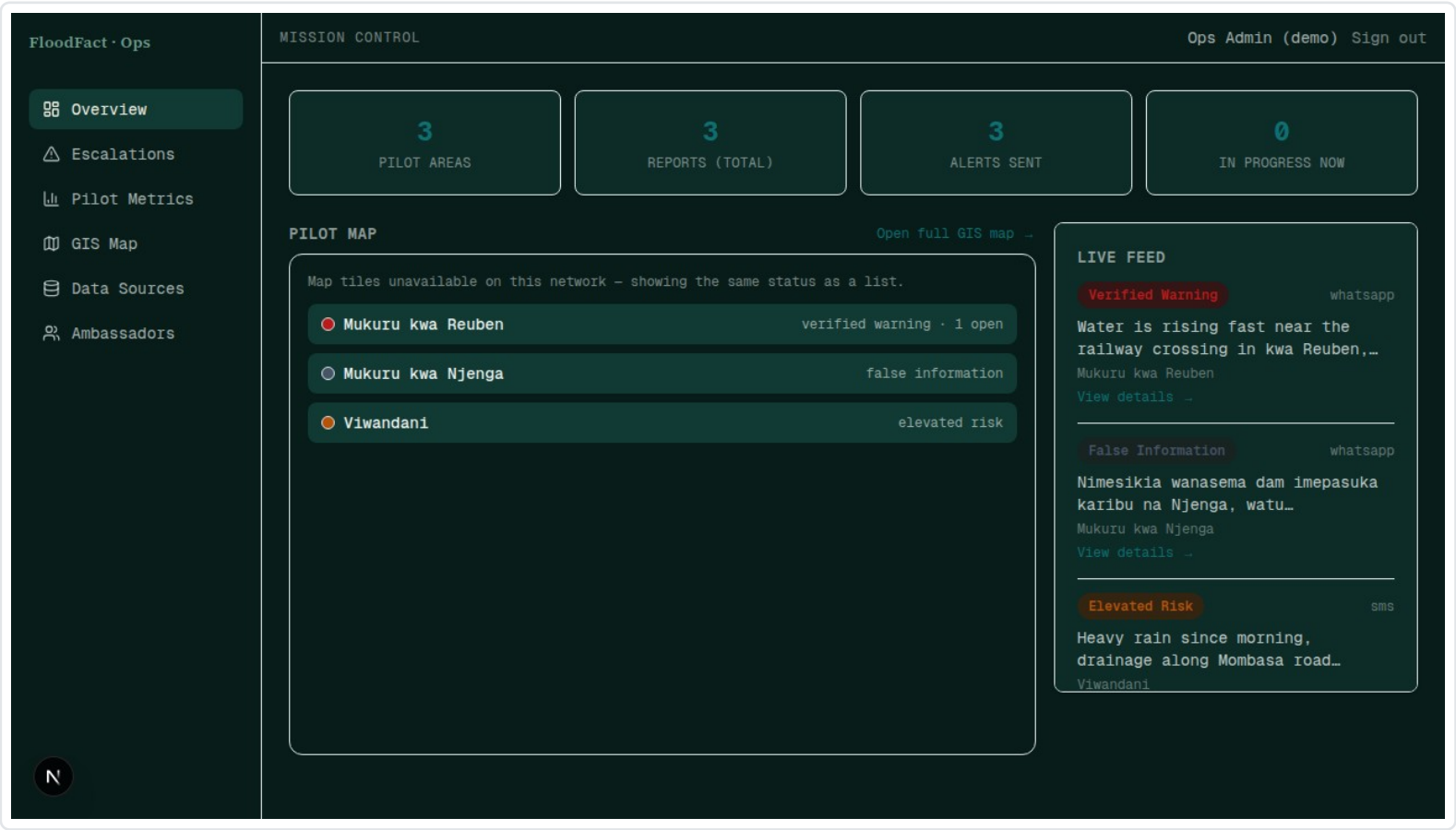
NOT JUST A PRODUCT

A real blog — the engineering rationale, not marketing copy



Six posts on how the pipeline works, why the AI doesn't decide, and general flood-safety guidance — no fabricated pilot results.

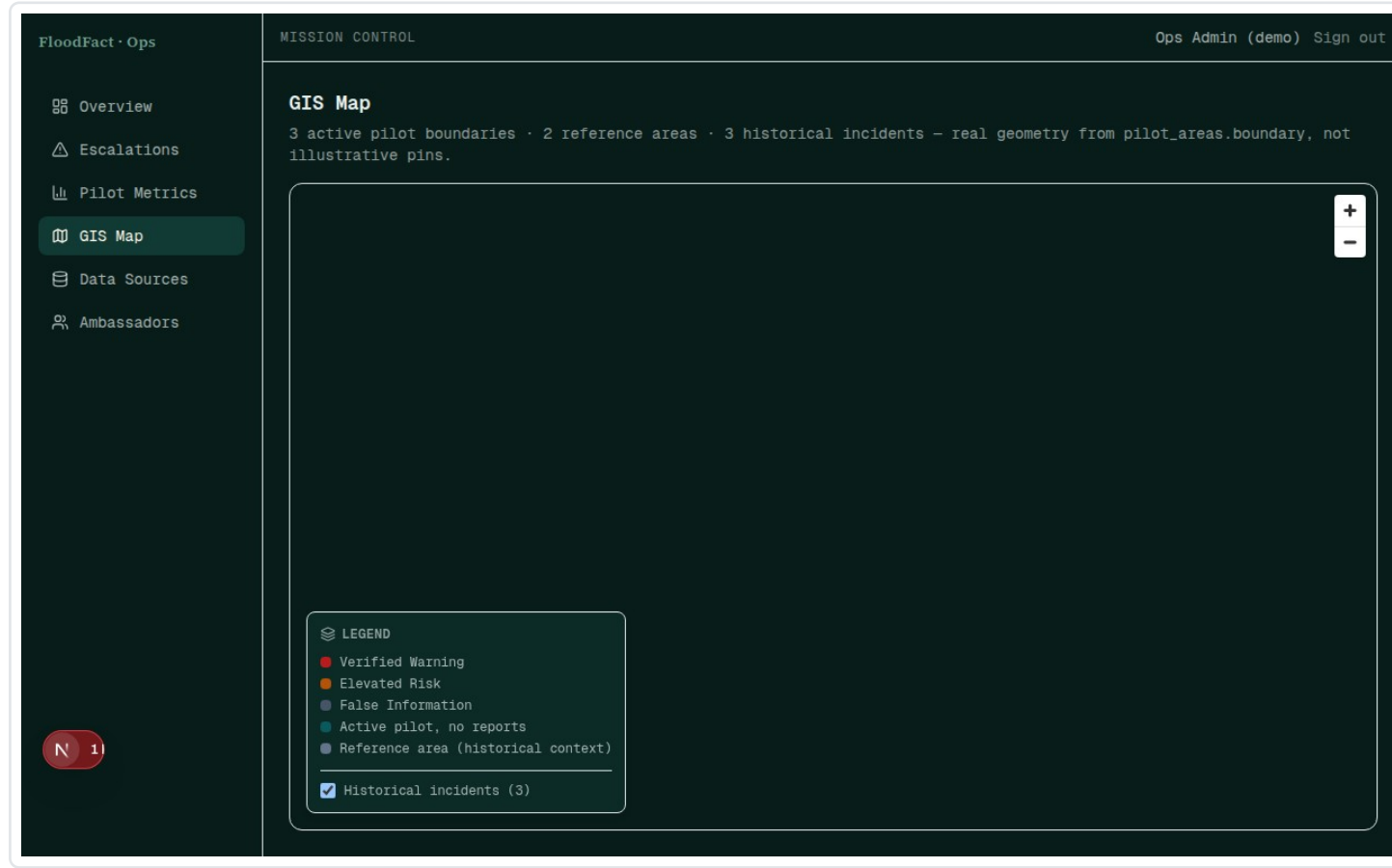
Every pilot area, one live map



Mukuru's three wards, colored by latest verdict — falls back to a status list if tiles don't load.

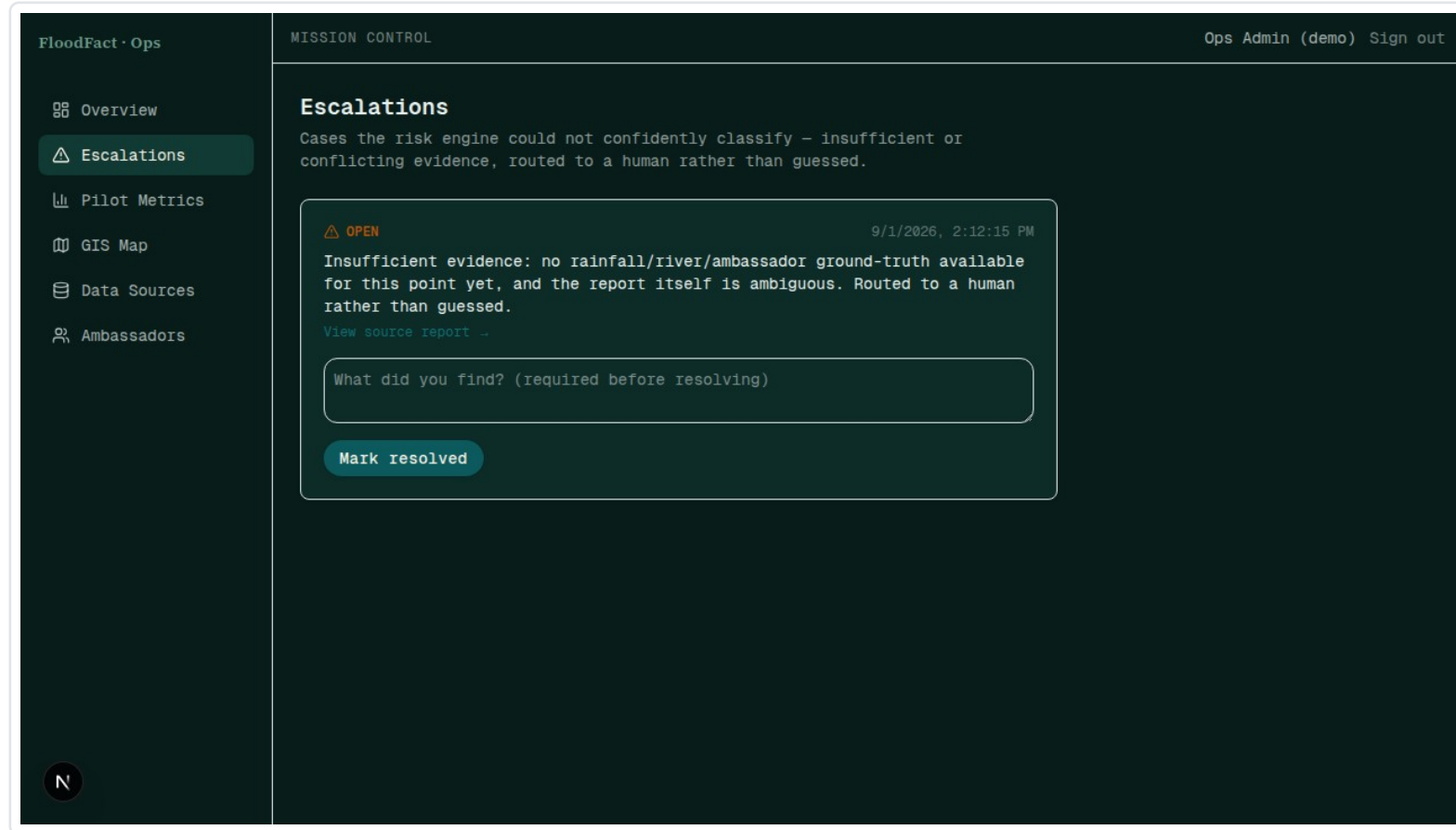
REAL GEOGRAPHY, NOT PINS

Pilot boundaries, reference wards, and historical incidents — one map



Real polygons from pilot_areas.boundary, plus cited historical floods — including an honestly-labeled illustrative one.

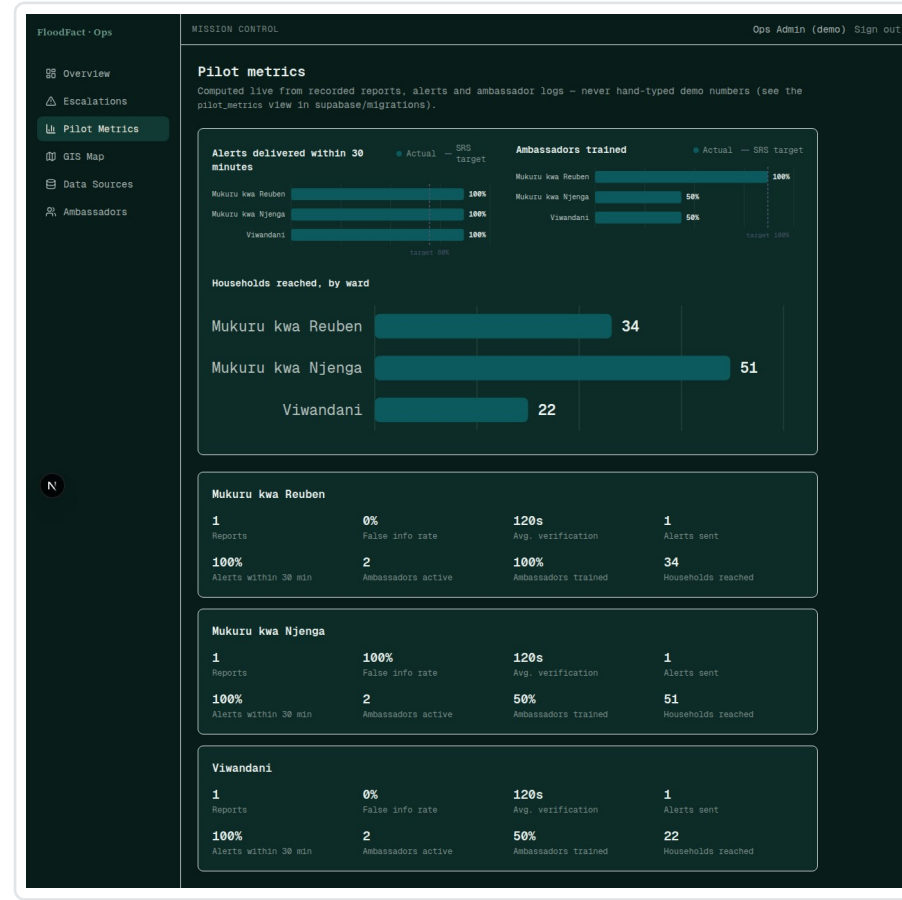
When evidence is insufficient, a human decides — never an algorithm alone



insufficient_evidence can never silently become “False Information” — it lands here for a human call.

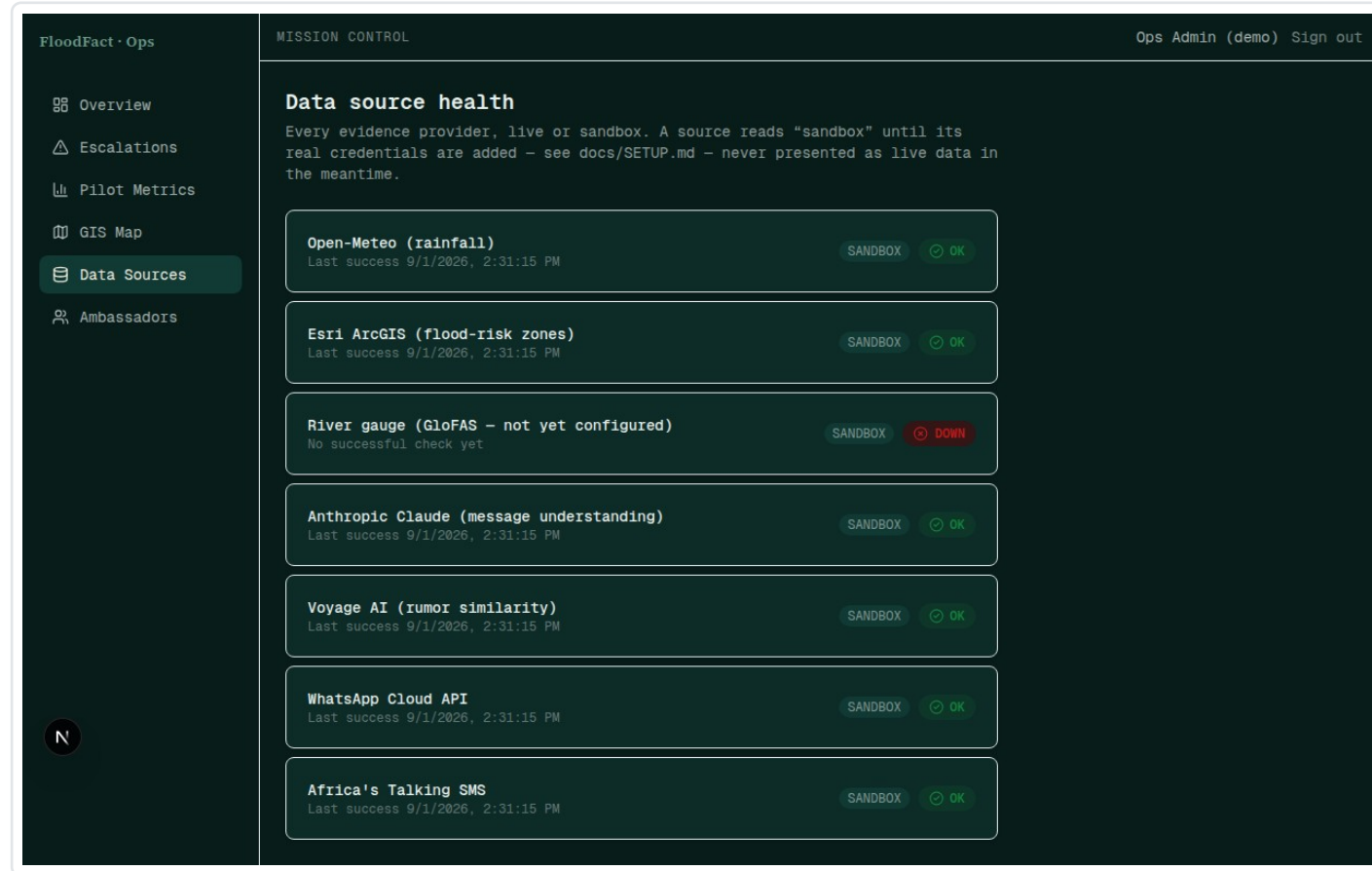
PROOF, NOT PROMISES

Pilot metrics computed from real database rows, not typed into a slide



A live SQL view, not hand-typed numbers — recomputed from reports and ambassador observations.

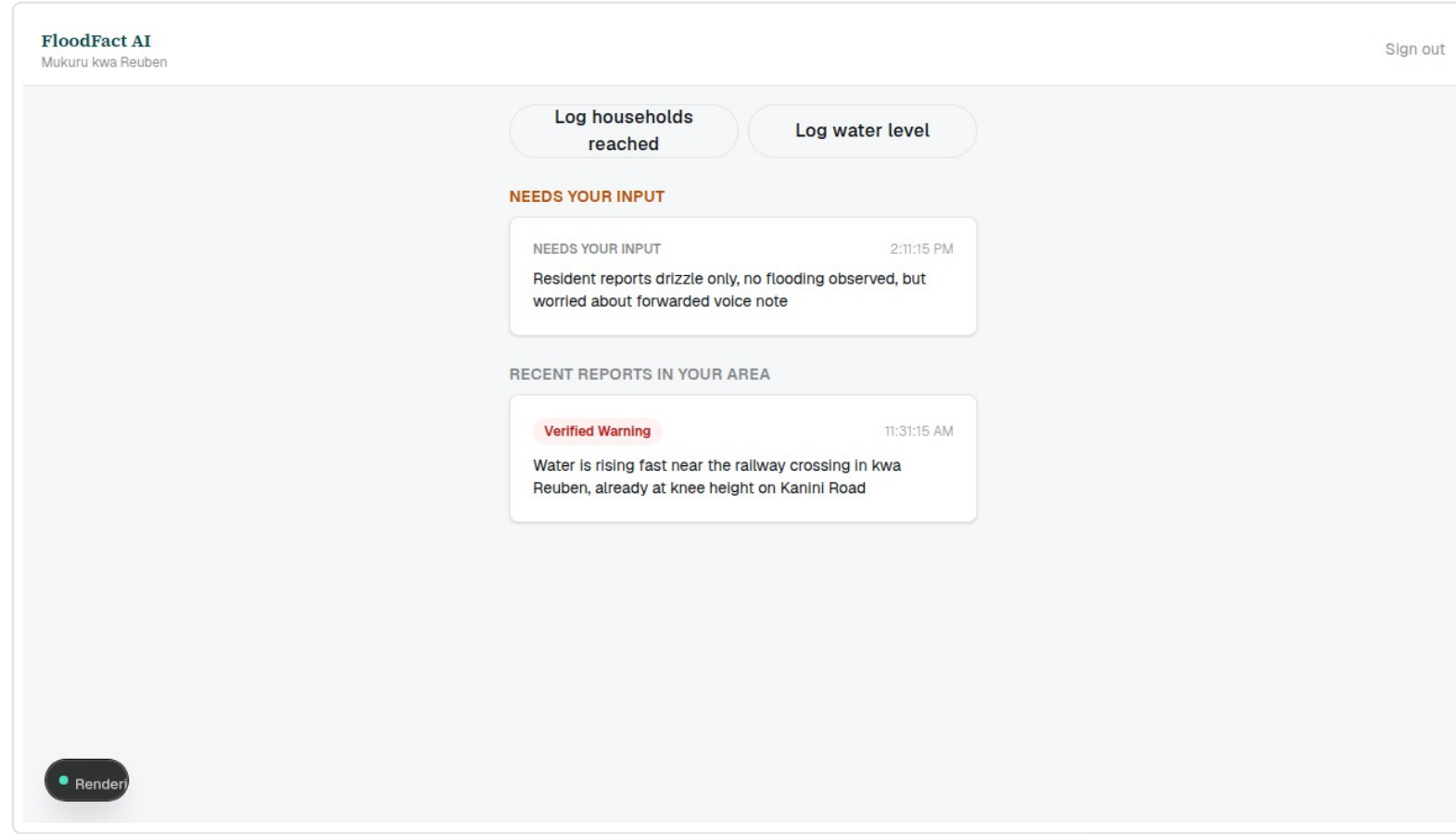
Every data source's health, live or sandbox, in plain sight



Each provider's live/sandbox mode and status, visible to the people operating the system.

STEP 6 — THE HUMAN TRUST LAYER

Youth ambassadors see only their own ward's queue



Row-level security enforces this at the database — an ambassador's session cannot query another ward.

A visual confirmation and a water-level reading feed straight back into the evidence store

FloodFact AI

Mukuru kwa Reuben

Sign out

← Back to queue

9/1/2026, 11:31:15 AM

Water is rising fast near the railway crossing in
kwa Reuben, already at knee height on Kanini
Road

Mukuru kwa Reuben

Verified Warning

Heavy rainfall in the last 3 hours, the location sits inside a mapped high flood-risk zone, similar events have happened here before, and 2 other nearby reports corroborate rising water.

Did you check this in person?

Yes, flooding confirmed

No flooding seen

N 1 Is

A visual confirmation and a water reading feed back in as evidence for the next nearby report.

Every integration switches independently — no code changes either way

Integration	Right now (zero paid keys)	Add one credential →
Rainfall & forecast	Open-Meteo — the real public API, already live (keyless)	No key needed — already real
Flood-risk zones	Seeded PostGIS polygon for Mukuru	Esri ArcGIS REST feature service
WhatsApp intake/alerts	Deterministic sandbox responder	Meta WhatsApp Cloud API (free test #)
SMS intake/alerts	Deterministic sandbox responder	Africa's Talking (Kenya)
Claim extraction + rationale	Deterministic template text	Anthropic Claude
Rumor-pattern similarity	Local hash embedding	Voyage AI + pgvector
Database, Auth, this deck	Static fixtures — DEMO_MODE, no Supabase project	Supabase Postgres + PostGIS + RLS

Each row switches on its own credential's presence, not a global flag. Exact steps for every row: docs/SETUP.md.

The whole system is one clone away

Read the code

Full source, 6 SQL migrations with RLS, and 43 passing unit tests — 23 of them exhaustively covering the risk engine's safety rules.

Run it yourself

`npm install && npm run dev` boots the entire product — public site, admin console, ambassador PWA — from seed fixtures. No accounts, no keys.

Go live

`docs/SETUP.md` walks through provisioning Supabase, Meta, Africa's Talking, Anthropic, and Voyage, step by step.

THANK YOU

**This is not a concept.
It's a working system, waiting on real
credentials.**

Concept: FloodFact-AI-Concept.pdf · Pilot Plan: FloodFact-AI-Pilot-Plan.pdf · Repository:
github.com/VinceOdundo/FloodfactAI